

# Case Study: E-Commerce — Spring Core + JdbcTemplate

## Objective

Build a simple e-commerce system using Spring's `JdbcTemplate`. Build DB connectivity using (HikariCP, DataSource config)

## Model classes:

**Category:** `id, name, description`

**Vendor:** `id, name, email`

**Product:** `id, name, price, stockQuantity, category_id (FK → Category), vendor_id (FK → Vendor)`

## Tasks

### Task 1 — Insert Product with FK references

Implement `save(Product product)` in `ProductRepository`, inserting `category_id` and `vendor_id` correctly.

### Task 2 — Find Product by ID with joined data

Implement `findById(Long id)` returning a `Product` (or DTO) that includes the category name and vendor name — using a `JOIN` query and a `RowMapper`.

### Task 3 — Update stock quantity

Implement `updateStock(Long productId, int newQuantity)` as a targeted `UPDATE`

### Task 4 — Aggregate query

Implement `countProductsByVendor()` returning a `Map<String, Integer>` (vendor name → product count) using `JdbcTemplate.query` with a `GROUP BY + JOIN`, and a `RowMapper` to build the map.